

ECON 101—Introduction to Microeconomics
Fall 2024

Final Exam

November 28, 2024

General Instructions

- Answer all questions. **All answers should be written in the answer sheet.** You can use as many scratch paper for your preliminary calculations as well as the exam booklet, but that work will not be considered. I will only consider the answers written in the answer sheet.
- At the end of the exam, you can take this booklet with you.
- The exam has a duration of 120 minutes.
- You can request a bathroom break, but only one person can be out at any given time.
- On your desk, you are allowed to have your ID, writing instruments, a ruler, and a simple calculator (no graphing calculators). You may also have a simple watch if needed (smartwatches are not allowed).
- Please place all other personal items, including smart devices, to the side of the room.
- If you complete the exam early, you may leave the room. but only within the first 60 minutes. Otherwise, please remain in the room quietly until the exam concludes.
- Crying is allowed, but please do so quietly to avoid disturbing others.
- Do not use the exam paper to wipe away tears.
- Good luck and may your choice of inputs maximize your profits in this exam!

Academic Dishonesty Policy and Guidelines at the End of the Exam

- Any form of cheating will be met with the **harshest consequences** available for the level of the offense. There will be **no sympathy** and **no leniency**.
- At the end of the exam, I will announce that the time is up. You **must follow these instructions**:
 - Put all your writing instruments down and close your exam booklet.
 - Keep your hands away from any writing instruments and your exam booklet.
 - Remain quiet and orderly. You are not free to leave until instructed to do so.
 - Any deviation from these rules will raise suspicion of academic dishonesty, and **you will be penalized accordingly**.

How to fill the bubbles and total points.

- Please use the answer sheet to record your answers. Only the answers in the answer sheet will be considered!
- Please use **only the provided pencil** to fill the bubbles! If you make a mistake be sure to erase your selection properly with the provided eraser!
- Examples:
 - Example: Correct answer. Q.1. ☒ (B) (C) (D)
 - Example: Incorrect way to mark an answer. Q.2. ☒ (B) (C) (D)
 - Example: Answer marked wrong because two bubbles are filled. Q.3. ☒ ☒ (C) (D)
- The exam has 20 questions worth 5 points each for a total of 100 points.

Please record all answers in the answer sheet. Each question is worth 5 points

Consider a firm with the following production function:

$$Q = A \cdot L^{0.5} K^{0.3} M^{0.2}$$

where:

- Q is the quantity of output,
- A is total factor productivity,
- L is the quantity of labor input,
- K is the quantity of capital input,
- M is the quantity of material input.

1. Suppose the firm increases all inputs by a constant factor $t > 1$. Which of the following statements correctly describes the returns to scale of this production function?
- (a) The production function exhibits **decreasing returns to scale**.
 - (b) The production function exhibits **variable returns to scale** depending on the value of t .
 - (c) The production function exhibits **increasing returns to scale**.
 - (d) The production function exhibits **constant returns to scale**.

Use the following information to answer questions 2–5. A profit-maximizing, price-taking firm produces output, widgets, y using a single input, labor, x . The wage rate of x is ₦450 per unit. The firm can produce 0, 8, 9, or 16 units of y using 0, 4, 7, or 9 units of x , respectively. There are no other input-output combinations. The firm production function is therefore given by:

$$y = f(x) = \begin{cases} 0 & \text{if } x = 0, \\ 8 & \text{if } x = 4, \\ 9 & \text{if } x = 7, \\ 16 & \text{if } x = 9. \end{cases}$$

2. What is the cost function $C(y)$ of this firm?

- (a) $C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 2,200 & \text{if } y = 8, \\ 3,150 & \text{if } y = 9, \\ 4,050 & \text{if } y = 16. \end{cases}$
- (b) $C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 2,000 & \text{if } y = 8, \\ 3,500 & \text{if } y = 9, \\ 4,050 & \text{if } y = 16. \end{cases}$
- (c) $C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 1,800 & \text{if } y = 8, \\ 3,150 & \text{if } y = 9, \\ 4,500 & \text{if } y = 16. \end{cases}$
- (d) $C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 1,800 & \text{if } y = 8, \\ 3,150 & \text{if } y = 9, \\ 4,050 & \text{if } y = 16. \end{cases}$

3. Let p be the price of a widget. What is the firm's supply function $y(p)$?

$$(a) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 8 & \text{if } 225 \leq p < 253.125, \\ 9 & \text{if } 253.125 \leq p < 315, \\ 16 & \text{if } p \geq 315. \end{cases}$$

$$(b) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 8 & \text{if } 225 \leq p < 253.125, \\ 9 & \text{if } 253.125 \leq p < 300, \\ 16 & \text{if } p \geq 300. \end{cases}$$

$$(c) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 200, \\ 8 & \text{if } 200 \leq p < 253.125, \\ 16 & \text{if } p \geq 253.125. \end{cases}$$

$$(d) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 8 & \text{if } 225 \leq p < 253.125, \\ 16 & \text{if } p \geq 253.125. \end{cases}$$

4. Suppose that there are 5 such price taking firms. What is the industry supply $Y^S(p)$?

$$(a) \ Y^S(p) = \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 40 & \text{if } 225 \leq p < 253.125, \\ 45 & \text{if } 253.125 \leq p < 315, \\ 80 & \text{if } p \geq 315. \end{cases}$$

$$(b) \ Y^S(p) = \begin{cases} 0 & \text{if } 0 \leq p < 200, \\ 40 & \text{if } 200 \leq p < 253.125, \\ 80 & \text{if } p \geq 253.125. \end{cases}$$

$$(c) \ Y^S(p) = \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 40 & \text{if } 225 \leq p < 253.125, \\ 45 & \text{if } 253.125 \leq p < 300, \\ 80 & \text{if } p \geq 300. \end{cases}$$

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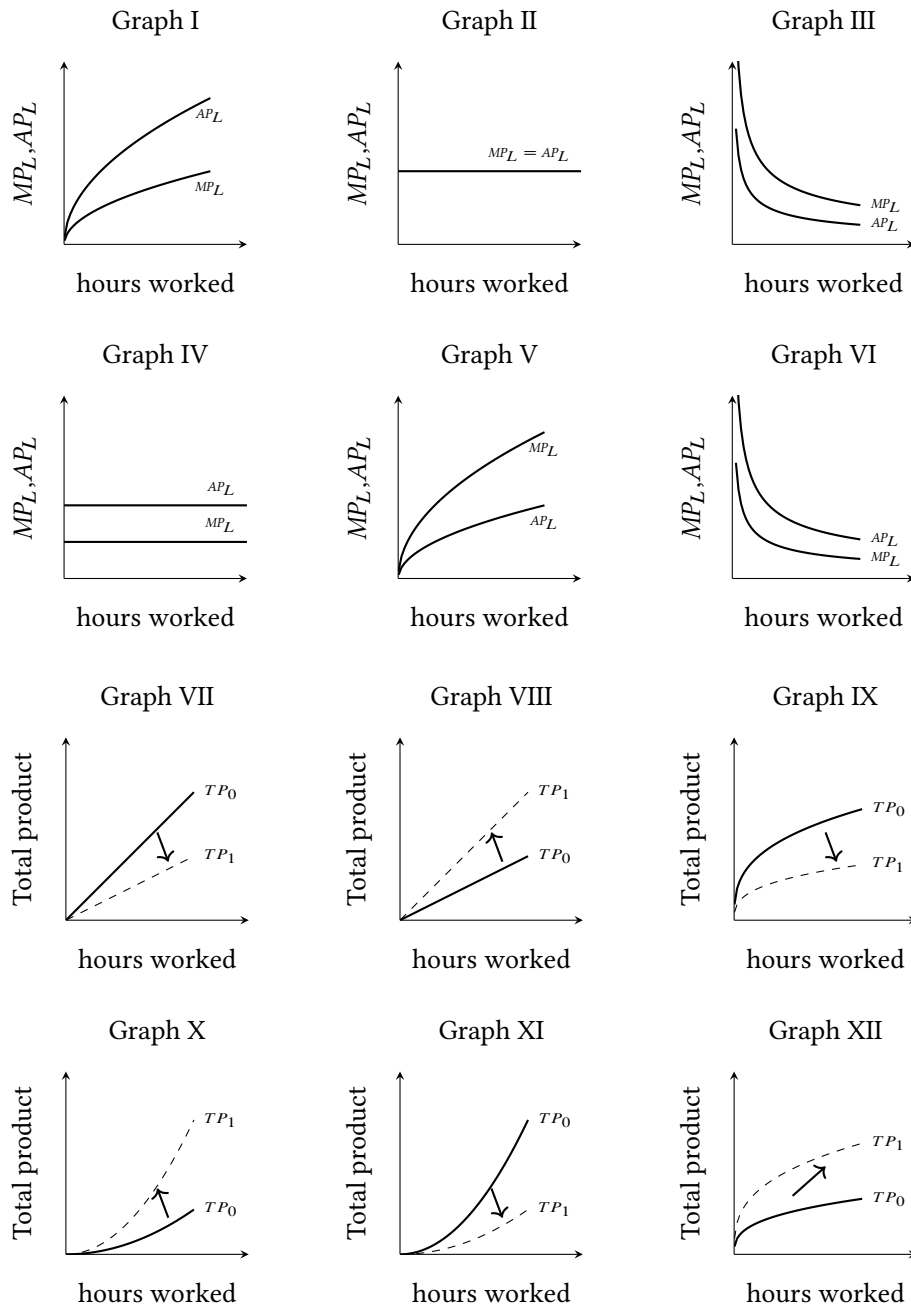
5. Suppose the market demand function is $Y^D(p) = 100 - \frac{p}{45}$. What is the market equilibrium?

- (a) $p^* = 900$; $Y^* = 80$ units
- (b) $p^* = 253.125$; $Y^* = 94.375$ units
- (c) $p^* = 225$; $Y^* = 95$ units
- (d) $p^* = 450$; $Y^* = 90$ units

Analyze the graphs below and answer questions 6 and 7.

Entrepreneur A claims that each additional worker they hire produces the same number of units as the previous worker. Entrepreneur B claims that each additional worker they hire produces fewer units than the previous worker. Entrepreneur C claims that each additional worker they hire produces more units than the immediately

preceding worker.



6. The graph _____ represents the average and marginal product of Entrepreneur A. The graph _____ represents the average and marginal product of Entrepreneur B. The graph _____ represents the average and marginal product of Entrepreneur C.

- (a) IV; VI; V.
- (b) I; II; III.
- (c) II; VI; V.
- (d) II; III; I.

7. If the three companies invest in a course on worker health care and on-the-job training, the graph _____ represents Company A, the graph _____ represents Company B, and the graph _____ represents Company C.

- (a) VIII; XII; X
- (b) VI; VII; XII
- (c) VII; IX; XI
- (d) VIII; X; XII

Use the following information to answer questions 8–11. Imagine the competitive plov market in Astana, where every vendor sells the most delicious plov. The market is **perfectly competitive** and has **free entry**, attracting anyone who dreams of becoming the “Plov King of Astana.”

Each vendor operates a small plov stall with the same *long-run cost structure*. The cost function for each vendor is:

$$C^A(q) = 49 + 5q + q^2, \text{ for } q > 0,$$

and if vendors decide not to produce, total cost is zero, i.e., $C^A(0) = 0$. All vendors sell plov at the same price p , as is common in perfectly competitive markets. Suppose that the market demand for plov is $D(p) = 235 - 5p$.

8. What is the vendor’s individual supply function?

- (a) $s(p) = \begin{cases} 0 & \text{if } p < 5, \\ \frac{p-5}{2} & \text{if } p \geq 5. \end{cases}$
- (b) $s(p) = \begin{cases} 0 & \text{if } p < 12, \\ \frac{p-5}{2} & \text{if } p \geq 12. \end{cases}$
- (c) $s(p) = \begin{cases} 0 & \text{if } p < 9, \\ \frac{p-5}{2} & \text{if } p \geq 9. \end{cases}$
- (d) $s(p) = \begin{cases} 0 & \text{if } p < 19, \\ \frac{p-5}{2} & \text{if } p \geq 19. \end{cases}$

9. What is the long-run market equilibrium price?

- (a) $p^* = 9$
- (b) $p^* = 19$
- (c) $p^* = 5$
- (d) $p^* = 12$

10. What is the long-run number of vendors in the market?

- (a) $N^* = 60$
- (b) $N^* = 20$
- (c) $N^* = 50$
- (d) $N^* = 70$

11. What is the quantity produced by each store in the long run?

- (a) $q^* = 5$
- (b) $q^* = 7$
- (c) $q^* = 6$
- (d) $q^* = 9$

12. A company is a price taker in the product market as well as in the input markets. Currently, the company employs 100 hours of labor at a wage of 50 Tenge per hour and uses 50 units of capital at a cost of 100 Tenge per unit. At the company's current level of production, the marginal product of labor is equal to 3, and the marginal product of capital is equal to 5. This company:
- (a) Could reduce its costs of producing this quantity by using less labor and more capital.
 - (b) Could reduce its costs of producing this quantity by using less capital and more labor.
 - (c) Could increase its production without incurring additional costs by using more capital and less labor.
 - (d) Is minimizing its costs for the quantity it produces.

Consider the following two statements:

Statement 1: Consider the market for cars. All firms have the same production technology, and we know that the long-run average cost curve has a minimum at a cost of 4,600,000 Tenge per car. Therefore, we can never observe a market equilibrium in which the equilibrium price is below 4,600,000 Tenge per car.

Statement 2: Consider a firm that produces computers. Suppose that the market price of a computer is $\bar{p}$, both in the short run and the long run. If in the short run this firm decides to shut down, then it must be that in the long run it decides to exit the market.

13. Which of the following options correctly describes the truth values of the statements?
- (a) Statement 1 is **False**; Statement 2 is **True**.
 - (b) Statement 1 is **True**; Statement 2 is **False**.
 - (c) Both statements are **True**.
 - (d) Both statements are **False**.

Consider an industry that has free entry and exit. Suppose that initially it is in a long-run equilibrium with constant costs with a fixed number of firms, and zero profits. Furthermore, suppose that firms are identical, and that they have upward sloping marginal costs. With this information, answer questions 14–17.

14. In the short-run, with a fixed number of firms, the supply curve is:
- (a) Vertical and perfectly inelastic
 - (b) Horizontal and perfectly elastic
 - (c) Upward sloping
 - (d) None of the above.
15. In the long-run, the long-run aggregate supply curve is:
- (a) Downward sloping and equal to the minimum average cost which is decreasing.
 - (b) Horizontal and equal to the minimum average cost.
 - (c) Upward sloping and equal to the minimum average cost which is increasing.
 - (d) I don't know and perhaps that is why I tried to pay someone to take the exam for me.
16. Now suppose that there is an increase in demand for the good that this industry produces. Then after the adjustment to the new equilibrium in the short-run:
- (a) The new equilibrium price will be higher compared to the original equilibrium price; producers will be making economic losses.
 - (b) The new equilibrium price will be lower compared to the original equilibrium price; producers will be making economic losses.
 - (c) The new equilibrium price will be higher compared to the original equilibrium price; producers will be making economic profits.
 - (d) The new equilibrium price will be lower compared to the original equilibrium price; producers will be making economic profits.

17. After the increase in the demand and the short-run adjustment, suppose that the government restricts entry into the industry by imposing a strict control on entry via operation permits. Then, in the long-run, the long-run aggregate supply is:
- Horizontal and equal to the minimum average cost.
 - Upward sloping due to the imposition of the operation permits.
 - Downward sloping due to the imposition of the operation permits.
 - I don't know because instead of studying I was trying to find someone who could take the exam for me.
18. The company “**Procrastination Bros.**” produces **custom meme hoodies**. If it hires 20 workers, it can produce 100 hoodies in one week. If it hires 35 workers, it can still produce 100 hoodies in one week. We can state that:
- If it hires 20 workers, the Marginal Product is positive and equal to the Average Product.
 - If it hires 20 workers, the Total Product is at its maximum.
 - If it hires 35 workers, the Marginal Product is negative and the Average Product is negative.
 - If it hires 35 workers, the Marginal Product is non-positive, but the Average Product is positive.

A semiconductor integrated circuit, or “chips” are the “brains” of computers and other electronic devices. Semiconductor manufacturers buy silicon wafers and then use labor and capital to produce the chips. A chip consists of multiple layers of silicon layers. A key step in the production process is to line up these layers. Suppose that there are two alternative alignment technologies that are available, each using different combinations of labor and capital:

Technology 1: Aligners Employees use machines, called *aligners*, which require workers to look through microscopes and line up the layers by hand. This is the least capital intensive technology. To produce 200 ten-layer chips per day takes 5 workers using 5 aligners.

Technology 2: Steppers Employees use a machine, called a *stepper*, which aligns the layers automatically. This technology requires less labor: To produce 200 ten-layer chips per day requires 3 workers and 6 steppers.

Technology 3: Stepper with machine wafer-handling In this technology, 2 workers, using 8 steppers with wafer-handling capabilities, can manufacture 200 ten-layer chips per day. This is the most capital-intensive technology.

Let L be workers per day, and K be units of capital (machines) used in each productive process. With the above information, answer questions 19 and 20.

19. Let F_1 , F_2 , and F_3 be the production functions describing each of the three technologies above respectively. Then:
- $F_1(L, K) = \min\{40L, 40K\}$, $F_2(L, K) = 200 \min\{\frac{L}{3}, \frac{K}{6}\}$, $F_3(L, K) = \min\{100L, 25K\}$.
 - $F_1(L, K) = \min\{40L, 40K\}$, $F_2(L, K) = \min\{\frac{200L}{3}, \frac{200K}{6}\}$, $F_3(L, K) = \min\{50L, 50K\}$.
 - $F_1(L, K) = \min\{40L, 40K\}$, $F_2(L, K) = \min\{\frac{200L}{6}, \frac{200K}{3}\}$, $F_3(L, K) = \min\{100L, 50K\}$.
 - $F_1(L, K) = \min\{40L, 40K\}$, $F_2(L, K) = 200 \min\{\frac{L}{6}, \frac{K}{3}\}$, $F_3(L, K) = \min\{50L, 50K\}$.
20. If the firm can freely choose what technological process to use, which verbal description below represents the correct description of the combined isoquant for producing 200 ten-layer chips when the three technologies are available.
- Three disconnected L-shaped isoquants for each technology, without any connection.
 - A single L-shaped isoquant at (2, 8), representing Technology 3 only.
 - A smooth convex curve connecting the points (5, 5), (3, 6), and (2, 8).
 - A piecewise-linear, convex curve with kinks at (5, 5), (3, 6), and (2, 8), formed by connecting the corners of the L-shaped isoquants.